LAB # 8

To Trace Different Traffic Flows in Computer Networks using Standard ACL

Objectives

- To follow the DHCP protocol configuration for IP address assignment using Packet Tracer
- To identify and understand configurations of standard ACL using Packet Tracer

Pre-Lab Exercise

Read this experiment in its entirety to become familiar with objectives of this lab. Study in detail and become familiar with the basics of Dynamic Host Configuration Protocol and Access Control Lists (ACLs) provided with this laboratory experiment. You may record the terms and sections that require more elaboration for reference. The instructor may provide the class some time to reflect upon these before proceeding with the lab.

Scenario A: Dynamic Host Configuration Protocol

The **Dynamic Host Configuration Protocol** (**DHCP**) is a standardized networking protocol used on IP networks that dynamically configures IP addresses and other information that is needed for Internet communication. DHCP allows computers and other devices to receive an IP address automatically from a central DHCP server, reducing the need for a network administrator or a user from having to configure these settings manually.

There are two methods to implement DHCP.

1. Router act as DHCP server and assign IPs to devices.
2. Router forward the DHCP requests to DHCP server.

Following Commands are used to implement DHCP protocol on Cisco Router.

R1(config)#ipdhcp pool *Pool_Name*

R1(dhcp-config)#network *Network ID Subnet Mask*

R1(dhcp-config)#default-router *Gateway IP*

R1(dhcp-config)#exit

Pool_Nmae: It is the pool name we can use whatever we want. This command gets us into the DHCP Configuration mode.

Network ID: It defines the network to be assigned IP's automatically.

Subnet Mask: It defines the network range to be leased.

Gateway IP: Default gateway for this network

Scenario B: Access Control List

Network security is a huge subject. One of the most important skills a network administrator needs is mastery of access control lists (ACLs). Administrators use ACLs to stop traffic or permit only specified traffic while stopping all other traffic on their networks.

Network designers use firewalls to protect networks from unauthorized use. Firewalls are hardware or software solutions that enforce network security policies. Consider a lock on a door to a room inside a building. The lock only allows authorized users with a key or access card to pass through the door. Similarly, a firewall filters unauthorized or potentially dangerous packets from entering the network. On a Cisco router, you can configure a simple firewall that provides basic traffic filtering capabilities using ACLs.

An ACL is a sequential list of permit or deny statements that apply to addresses or upper-layer protocols. ACLs provide a powerful way to control traffic into and out of your network. You can configure ACLs for all routed network protocols.

What is an ACL?

An ACL is a router configuration script that controls whether a router permits or denies packets to pass based on criteria found in the packet header. ACLs are also used for selecting types of traffic to be analyzed, forwarded, or processed in other ways. You can configure one ACL per protocol, per direction, per interface:

- One ACL per protocol-To control traffic flow on an interface, an ACL must be defined for each protocol enabled on the interface.
- One ACL per direction-ACLs control traffic in one direction at a time on an interface. Two separate ACLs must be created to control inbound and outbound traffic.
- One ACL per interface-ACLs control traffic for an interface, for example, Fast Ethernet 0/0.

ACL Operation

How ACLs Work

ACLs define the set of rules that give added control for packets that enter inbound interfaces, packets that relay through the router and packets that exit outbound interfaces of the router. ACLs do not act on packets that originate from the router itself.

ACLs are configured either to apply to inbound traffic or to apply to outbound traffic.

Inbound ACLs-Incoming packets are processed before they are routed to the outbound interface. An inbound ACL is efficient because it saves the overhead of routing lookups if the packet is discarded. If the packet is permitted by the tests, it is then processed for routing.

Outbound ACLs-Incoming packets are routed to the outbound interface, and then they are processed through the outbound ACL.

ACL statements operate in sequential order. They evaluate packets against the ACL, from the top down, one statement at a time. If a packet header and an ACL statement match, the rest of the statements in the list are skipped, and the packet is permitted or denied as determined by the matched statement. If a packet header does not match an ACL statement, the packet is tested against the next statement in the list. This matching process continues until the end of the list is reached.

Access List Types

Cisco IOS access lists are divided into two distinct types:

- **Standard ACLs:** This type of ACL is the simplest one since it only filters traffic based on source IP addresses. In other words, this ACL can be used only when you need to permit or deny traffic from a specific host IP address or a specific source network.
- **Extended ACLs:** This type of ACL is the most preferred one and the most advanced as well. Extended ACLs filter IP packets based on several attributes, for example, protocol type, source IP address, destination IP address, source TCP or UDP ports, destination TCP or UDP ports, and optional protocol type information for finer granularity of control.

Numbering and Naming ACLs

Using numbered ACLs is an effective method for determining the ACL type on smaller networks with more homogeneously defined traffic. However, a number does not inform you of the purpose of the ACL. For this reason you can use a name to identify a Cisco ACL.

The rule to designate numbered ACLs and named ACLs is:

Numbered ACL

You assign a number based on which protocol you want to filtered:

- (1 to 99) and (1300 to 1999): Standard IP ACL
- (100 to 199) and (2000 to 2699): Extended IP ACL

In case you are wondering why numbers 200 to 1299 are skipped, it is because those numbers are used by other protocols.

Named ACL

You assign a name by providing the name of the ACL:

- Names can contain alphanumeric characters.
- It is suggested that the name be written in CAPITAL LETTERS.
- Names cannot contain spaces or punctuation and must begin with a letter.
- You can add or delete entries within the ACL.

Where to Place ACLs

The proper placement of an ACL to filter undesirable traffic makes the network operate more efficiently. ACLs can act as firewalls to filter packets and eliminate unwanted traffic. Where you place ACLs can reduce unnecessary traffic. For example, traffic that will be denied at a remote destination should not use network resources along the route to that destination.

Every ACL should be placed where it has the greatest impact on efficiency. The basic rules are:

- Because standard ACLs do not specify destination addresses, place them as close to the destination as possible.
- Locate extended ACLs as close as possible to the source of the traffic denied. This way, undesirable traffic is filtered without crossing the network infrastructure.

In-Lab Exercise

In this lab activity, you will create a network that is similar to the one shown in the Topology Diagram. Begin by cabling the network as shown in the Figure 15. Assign Network_ID from Class C as shown in the figure and fill in the table. You will then perform the initial router configurations required for connectivity. Use the IP addresses according to your IP assignment. First test the connections between directly connected devices, and then test connectivity between devices that are not directly connected. You will configure the dynamic routing protocol RIP to add remote networks, this will allow communication between the hosts of different networks. View the routing table after each dynamic route is added to observe how the routing table has changed. After that, configure DHCP protocol on different networks so that all hosts from the network get IP address dynamically. Also create Access Control Lists (ACLs) to block networks from gaining access according to the given scenarios.

Topology Diagram

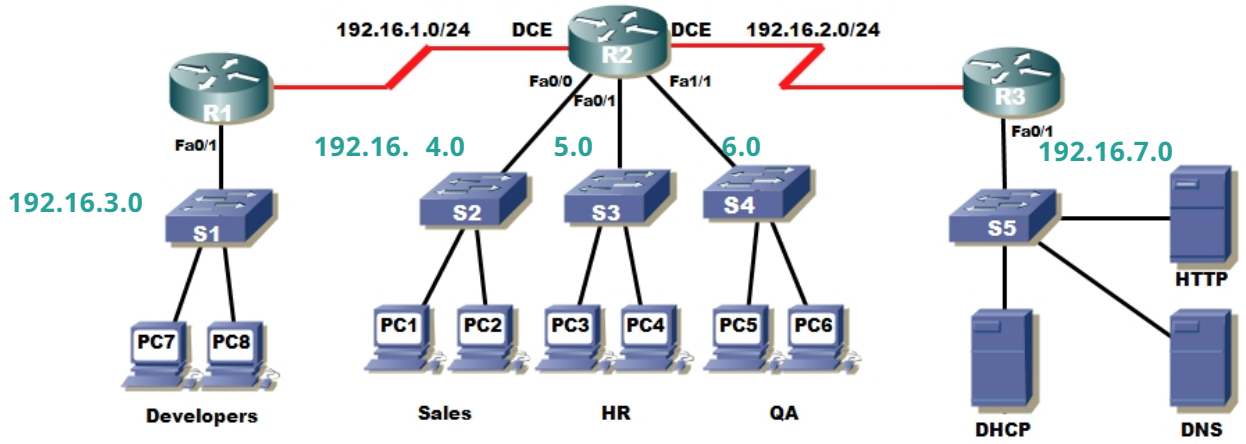

Figure 15: Network Topology

Addressing Table

Device	Interface	IP Address	Subnet Mask	Gateway
R1	Se0/1/0	192.16.1.1	255.255.255.0	
	Fa0/0	192.16.3.1	255.255.255.0	
R2	Se0/1/0	192.16.1.2	255.255.255.0	
	Se0/2/0	192.16.2.1	255.255.255.0	
	Fa0/0	192.16.4.1	255.255.255.0	
	Fa0/1	192.16.5.1	255.255.255.0	
	Fa1/0	192.16.6.1	255.255.255.0	
R3	Se0/2/0	192.16.2.2	255.255.255.0	
	Fa0/0	192.16.7.1	255.255.255.0	
HTTP Server	N/A	192.16.7.2	255.255.255.0	192.16.7.1
DNS Server	N/A	192.16.7.3	255.255.255.0	192.16.7.1
DHCP Server	N/A	192.16.7.4	255.255.255.0	192.16.7.1
PC1 – PC8	NIC	Obtain Automatically	Obtain Automatically	

Task 1: Prepare the Network

Task 2: Perform Basic Router Configurations.

Step 1: Configure Ethernet interfaces on R1.

Step 2: Verify IP addressing and interfaces.

Use the *show ip interface brief* command to verify that the IP addressing is correct and that the interfaces are active.

Task 3: Perform IP configuration on host PC's

Now you don't need to assign IP address on each PC. We will use DHCP to assign to get IP address automatically. Select DHCP on instead of Static on PC IP Configuration.

Step 1: Configure DHCP for Developer's Network

Complete the following commands and run on Router R1.

R1(config)# ip dhcp pool DEVELOPERS

R1(dhcp-config)# network 192.16.3.0 255.255.255.0

R1(dhcp-config)# default-router 192.16.3.1

R1(dhcp-config)# dns-server 192.16.7.3

Step 2: Configure DHCP for Sales Network

Complete the following commands and run on router R2.

R1(config)# ip dhcp pool SALES

R1(dhcp-config)# network 192.16.4.0 255.255.255.0

R1(dhcp-config)# default-router 192.16.4.1

R1(dhcp-config)# dns-server 192.16.7.3

Step 3: Configure DHCP for HR Network

Complete the following commands and run on router R2.

R1(config)# ip dhcp pool HR

R1(dhcp-config)# network 192.16.5.0 255.255.255.0

R1(dhcp-config)# default-router 192.16.5.1

R1(dhcp-config)# dns-server 192.16.7.3

Step 4: Configure DHCP for QA Network

Complete the following commands and run on router R2.

R1(config)# ip dhcp pool QA

R1(dhcp-config)# network 192.16.6.0 255.255.255.0

R1(dhcp-config)# default-router 192.16.6.1

R1(dhcp-config)# dns-server 192.16.7.3

Task 4: Verify the Network

Use ping tool to check connectivity between the hosts. Is ping successful?

yes

Task 5: Configuring Standard IP Access Lists

The access list may be created with one or more access-list commands while in global configuration mode. Second, the access list is applied to or referenced by other commands, such as the ip access-group command which applies the ACL to an interface. An example would be the following:

R1#config terminal

R1(config)#access-list ***# {permit | deny} ip address wildcard mask***

R1(config)#interface ***{serial | Fast Ethernet}***

R1(config-if)#ip access-group ***# {in | out}***

***Implicit deny statement*:** Every access list contains a final "deny" statement that matches all packets. This is called the implicit deny. Because the implicit deny statement is not visible in show command output, it is often overlooked, with serious consequences.

At least one permit statement is required: There is no requirement that an ACL contain a deny statement. If nothing else, the implicit deny any statement takes care of that. But if there are no permit statements, the effect will be the same as if there were only a single deny any statement.

Wildcard mask: In identifying IP addresses, ACLs use a wildcard mask instead of a subnet mask. Initially, the masks might look the same, but closer observation reveals that they are very different. Remember that a binary 0 in a wildcard mask instructs the router to match the corresponding bit in the IP address.

In/out: When deciding whether an ACL should be applied to inbound or outbound traffic, always view things from the perspective of the router. In other words, determine whether traffic is coming into the router, inbound, or leaving the router, outbound.

Applying ACLs: Extended ACLs should be applied as close to the source as possible, thereby conserving network resources. Standard ACLs, by necessity, must be applied as close to the destination as possible. This is because the standard ACL can match only at the source address of a packet.

Standard ACLs

Step 1: Your task is to block 192.16.3.3 from gaining access on 192.16.5.0. While 192.16.3.3 must be able to communicate with networks. Other computer from the network of 192.16.3.0 must be able to connect with the network of 192.16.5.0.

Run following commands on router R2.

R2#config terminal

R2(config)#access-list 1 deny host 192.16.3.3

R2(config)#access-list 1 permit any

R2(config)#interface fastEthernet 0/1

R2(config-if)#ip access-group 1 out

Step 2: Verify if the access list is created successfully.

Run following command on router R2 and write down your observations.

R2# ***show access-list***

```
Standard IP access list 1
    10 deny host 192.16.3.3
    20 permit any
```

Step 3: Block the network of 192.16.3.0 from gaining access on 192.16.4.0. While 192.16.3.0 must be able to communicate with networks.

Run following commands on router R2.

R2(config)#access-list 2 deny 192.16.3.0 0.0.0.255

R2(config)#access-list 2 permit any

R2(config)#interface fastethernet 0/0

R2(config-if)#ip access-group 2 out

Rubric for Lab Assessment

The student performance for the assigned task during the lab session was:			
Excellent	The student completed assigned tasks without any help from the instructor and showed the results appropriately.	4	
Good	The student completed assigned tasks with minimal help from the instructor and showed the results appropriately.	3	
Average	The student could not complete all assigned tasks and showed partial results.	2	
Worst	The student did not complete assigned tasks.	1	

Instructor Signature: _____ Date: _____